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1 Data Science

Reference basis: “Data Science Bootcamp Course from Udemy” - written/compiled by Maitreya Ranade.

2 Introduction

Please find the Big picture of Data science here in :

Some important things to keep in mind:

- **Analytics vs Analysis** Term Analytics relates to the future events. It includes exploring of patterns and prediction. Term analysis relates to the data from past events(existing data). It explains how and when.
- **Business intelligence** Process of analysing and reporting historical business data. Aims to explain past events using business data.
- **Machine learning** The ability of machines to predict outcomes without being explicitly programmed. Creating and implementing algorithms that let machines receive data and use this data to: 1. Make predictions, 2. Analyse Patterns, 3. Give recommendations.
- **Business intelligence** Simulating human knowledge and decision making with computers.

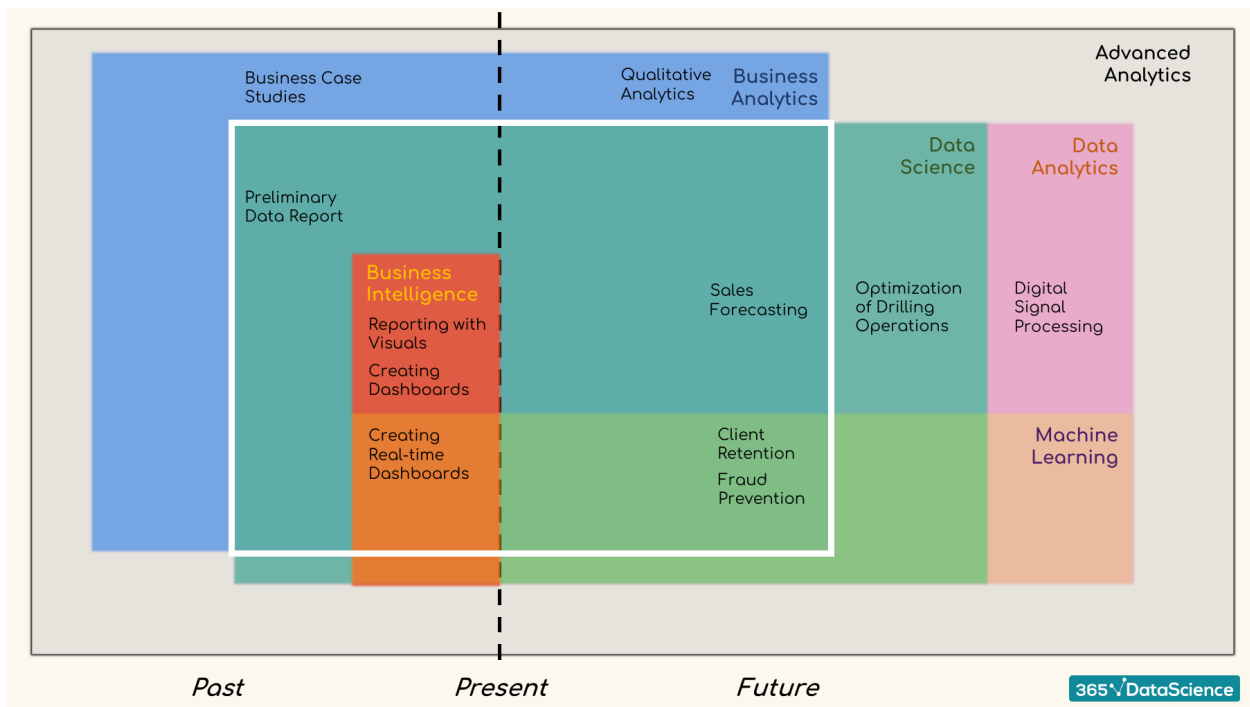


Figure 1: Big picture of the data science term definitions

3 Probability

3.1 Combinatorics

Permutations (Arrange)(Used for arrangement of objects & order is relevant): Arrange entire set of elements in the sample space. Variations (Pick and arrange) (Used for arrangement of objects & order is relevant): Arrange only a few elements from the set of elements in the sample space. Combination (Pick)(order is irrelevant)

No repetition

Repetition

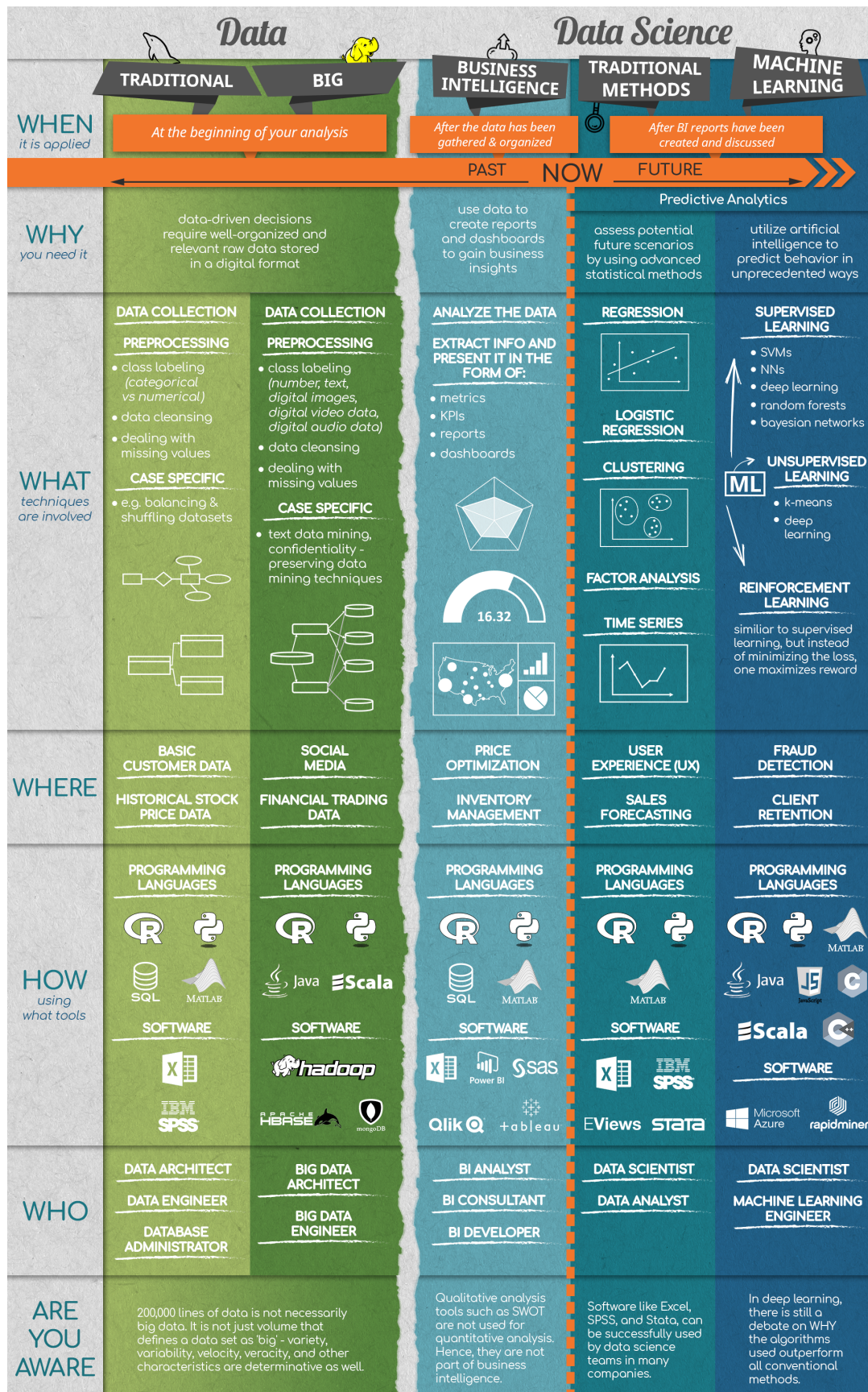


Figure 2: Overview of Course

3.2 Bayesian Inference

3.3 Distributions

3.4 Probability in Other Fields

4 Statistics

4.1 Descriptive Statistics

4.2 Practical Example: Descriptive Statistics

4.3 Inferential Statistics Fundamentals

4.4 Inferential Statistics: Confidence Intervals

4.5 Practical Example: Inferential Statistics

4.6 Hypothesis Testing

4.7 Practical Example: Hypothesis Testing

5 Introduction to Python

5.1 Variables and Data Types

5.2 Basic Python Syntax

5.3 Other Python Operators

5.4 Conditional Statements

5.5 Python Functions

5.6 Sequences

5.7 Iterations

5.8 Advanced Python Tools

6 Advanced Statistical Methods in Python

6.1 Linear regression with StatsModels

6.2 Multiple Linear Regression with StatsModels

6.3 Linear Regression with sklearn

6.4 Practical Example: Linear Regression

6.5 Logistic Regression

6.6 Cluster Analysis

6.7 K-Means Clustering

6.8 Other Types of Clustering

7 Mathematics

8 Deep Learning

8.1 Introduction to Neural Networks

8.2 How to Build a Neural Network from Scratch with NumPy